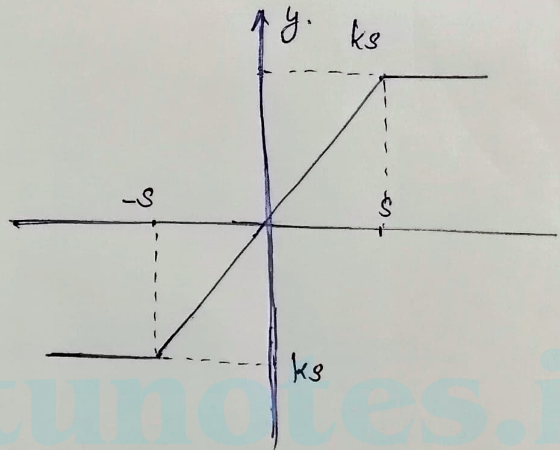
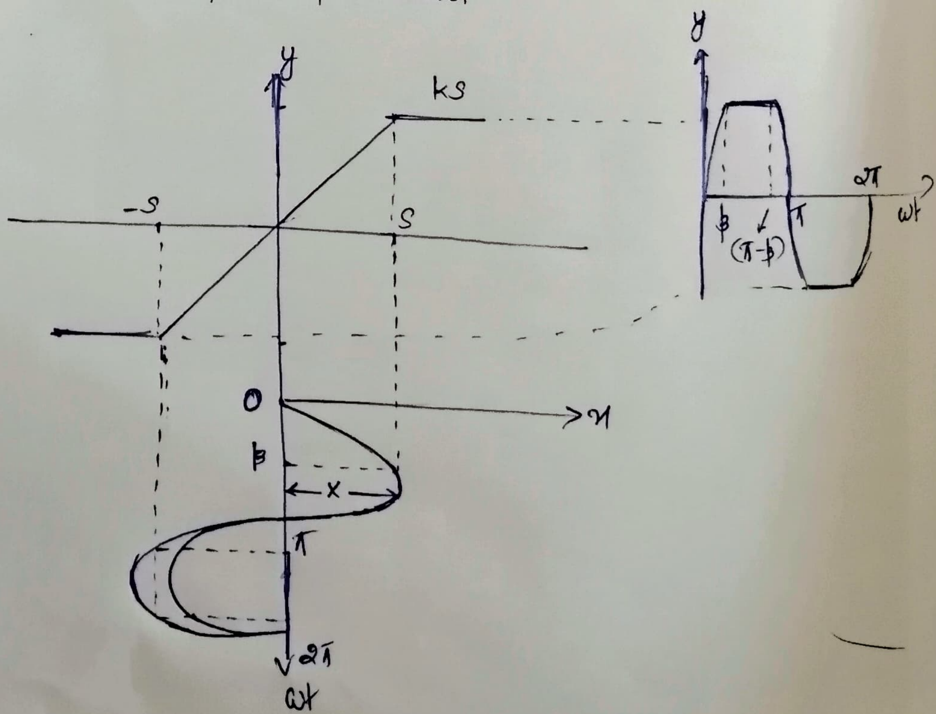
Analysis of Describing Method for Non-Linear System

Imp (Characteristics and types of non-linear s/m's)

Describing function for saturation non-linearity



Input output chara



The i/p - o/p relation is linear for $x = 0 \rightarrow S$ then when the i/p, $x > S$, the o/p reaches a saturated value of KS.

The i/p is sinusoidal $\therefore x = X \sin\omega t$ —— ①

when $\omega t = \beta$, $x = S$

$$\therefore ① \Rightarrow S = X \sin\beta.$$

$$\sin\beta = \frac{S}{X}$$

$$\beta = \sin^{-1}\left(\frac{S}{X}\right)$$

The o/p y of non-linearity is divided into 3 regions in the range from : $0 \rightarrow \pi$

$$y = Kx, \quad 0 \le \omega t \le \beta$$

$$= KS, \quad \beta \le \omega t \le \pi - \beta$$

$$= -Kx, \quad \pi - \beta \le \omega t \le \pi.$$

The describing function is given by,

$$K_N(X,\omega) = \frac{Y_1}{X} \angle \phi_1$$

where, $Y_1 = \sqrt{A_1^2 + B_1^2}$ and $\phi_1 = \tan^{-1}(B_1/A_1)$.

For half wave and quarter wave symmetry, $B_1 = 0$ and A_1 is given by.

$$A_1 = \frac{2}{\pi/2} \int_0^{\pi/2} y \sin\omega t \cdot d(\omega t).$$

$$= \frac{4}{\pi}\left[\int_0^{\beta} y \sin\omega t \, d(\omega t) + \int_{\beta}^{\pi/2} y \sin\omega t \, d(\omega t)\right]$$

$$= \frac{4}{\pi}\left[\int_0^{\beta} kx \cdot \sin\omega t \, d(\omega t) + \int_{\beta}^{\pi/2} ks \cdot \sin\omega t \, d(\omega t)\right]$$

$$= \frac{4}{\pi}\left[\int_0^{\beta} k \cdot X \sin\omega t \cdot \sin\omega t \, d(\omega t) + \int_{\beta}^{\pi/2} ks \cdot \sin\omega t \, d(\omega t)\right]$$

$$= \frac{4kX}{\pi}\int_0^{\beta} \sin^2\omega t \cdot d(\omega t) + \frac{4ks}{\pi}\int_{\beta}^{\pi/2} \sin\omega t \cdot d(\omega t).$$

$$= \frac{4kX}{\pi}\int_0^{\beta} \left(\frac{1-\cos 2\omega t}{2}\right) d(\omega t) + \frac{4ks}{\pi}\int_{\beta}^{\pi/2} \sin\omega t \cdot d(\omega t)$$

$$= \frac{4kX}{2\pi}\left[\omega t - \frac{\sin 2\omega t}{2}\right]_0^{\beta} + \frac{4ks}{\pi}\left[-\cos\omega t\right]_{\beta}^{\pi/2}$$

$$= \frac{2kX}{\pi}\left[\beta - \frac{\sin 2\beta}{2}\right] - \frac{4ks}{\pi}\left(\cos \pi/2 - \cos\beta\right)$$

$$= \frac{2kX}{\pi}\left[\beta - \frac{\sin 2\beta}{2}\right] + \frac{4ks}{\pi}\cos\beta.$$

Substituting for $S = X \sin\beta$

$$A_1 = \frac{2kX}{\pi}\left[\beta - \frac{\sin 2\beta}{2}\right] + \frac{4kX \sin\beta \cos\beta}{\pi}$$

$$= \frac{2kX}{\pi}\left[\beta - \frac{2\sin\beta\cos\beta}{2}\right] + \frac{4kX}{\pi}\sin\beta\cdot\cos\beta$$

$$= \frac{2kX}{\pi}\left[\beta - \cancel{2}\sin\beta\cos\beta + 2\sin\beta\cos\beta\right]$$

$$= \frac{2kX}{\pi}\left[\beta + \sin\beta\cos\beta\right]$$

$$\therefore Y_1 = \sqrt{A_1^2 + B_1^2} = \sqrt{A_1^2 + 0} = A_1$$

$$= \frac{2kX}{\pi}\left[\beta + \sin\beta\cos\beta\right]$$

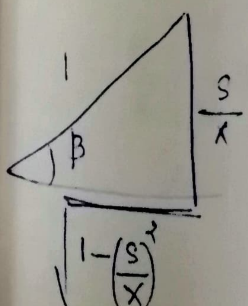
$$\phi_1 = \tan^{-1}\left(\frac{B_1}{A_1}\right) = \tan^{-1}(0) = \underline{\underline{0^\circ}}$$

The describing func, $K_N(X,\omega) = \frac{Y_1}{X}\angle\phi_1$

$$K_N(X,\omega) = \frac{Y_1}{X}\angle\phi_1 = \frac{2K}{\pi}(\beta + \sin\beta\cos\beta)\angle 0^\circ$$

We've, $\sin\beta = \frac{S}{X}$

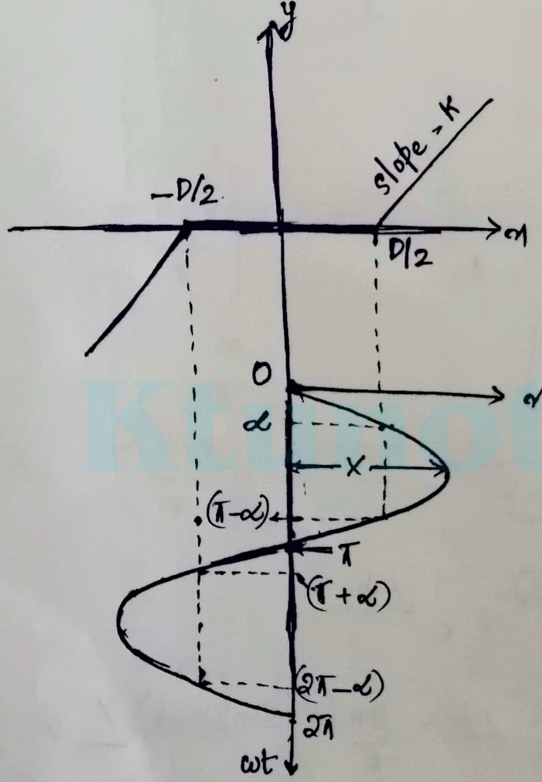
$$\cos\beta = \sqrt{1-\left(\frac{S}{X}\right)^2}$$

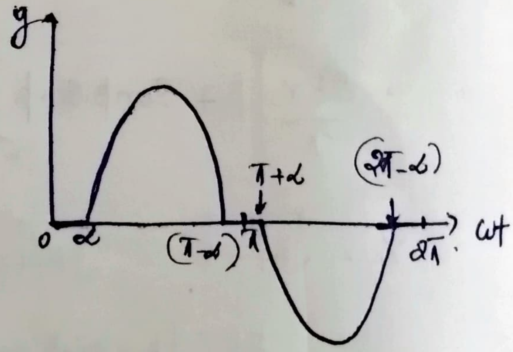


$$\therefore K_N(X,\omega) = \frac{2K}{\pi}\left[\beta + \sin\beta\cos\beta\right]$$

$$= \frac{2K}{X}\left[\sin^{-1}\left(\frac{S}{X}\right) + \frac{S}{X}\sqrt{1-\left(\frac{S}{X}\right)^2}\right]$$

Describing function of deadzone non linearity :





The o/p is 0 when the i/p is less than D/2. and i/p-o/p relation is linear when the i/p is > D/2.

When the i/p is sinusoidal, $x = X\sin\omega t$ —①

when $\omega t = \alpha$, $x = D/2$.

$$\therefore ① \Rightarrow \frac{D}{2} = X\sin\alpha$$

$$\sin\alpha = \frac{D}{2X}$$

$$\alpha = \sin^{-1}\left(\frac{D}{2X}\right)$$

The o/p is divided into 3 regions

$$y = \begin{cases} 0, & 0 \le \omega t \le \alpha \\ K\left(A - \frac{D}{2}\right), & \alpha \le \omega t \le \pi - \alpha \\ 0, & \pi - \alpha \le \omega t \le \pi \end{cases}$$

Describing function is given by,

$$K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1$$

where, $Y_1 = \sqrt{A_1^2 + B_1^2}$ and $\phi_1 = \tan^{-1}\left(\frac{B_1}{A_1}\right)$.

The o/p has half wave and ~~squa~~ quarter wave symmetry,

$B_1 = 0$.

$\therefore Y_1 = \sqrt{A_1^2} = A_1$.

$$A_1 = \frac{2}{\pi/2}\int_0^{\pi/2} y \sin \omega t \, d(\omega t).$$

$$= \frac{4}{\pi}\int_0^{\pi/2} y \sin \omega t \, d(\omega t).$$

$$= \frac{4}{\pi}\left[\int_0^{\alpha} y \sin \omega t \, d(\omega t) + \int_{\alpha}^{\pi/2} y \sin \omega t \, d(\omega t)\right]$$

$$= \frac{4}{\pi}\int_{\alpha}^{\pi/2} y \sin \omega t \cdot d(\omega t).$$

$$= \frac{4}{\pi}\int_{\alpha}^{\pi/2} K\left(x-\frac{D}{2}\right) \sin\omega t \, d(\omega t)$$

$$= \frac{4}{\pi}\int_{\alpha}^{\pi/2} K\left(X\sin\omega t-\frac{D}{2}\right) \sin\omega t \, d(\omega t)$$

$$= \frac{4}{\pi}\left[\int_{\alpha}^{\pi/2} KX\sin^2\omega t \, d(\omega t) - \int_{\alpha}^{\pi/2} \frac{KD}{2}\sin\omega t \, d(\omega t)\right]$$

$$= \frac{4}{\pi}\left[\int_{\alpha}^{\pi/2} KX\left(\frac{1-\cos 2\omega t}{2}\right) d(\omega t) - \int_{\alpha}^{\pi/2} \frac{KD}{2}\sin\omega t \, d(\omega t)\right]$$

$$= \frac{4K}{\pi}\left[\frac{X}{2}\int_{\alpha}^{\pi/2}(1-\cos 2\omega t)\, d(\omega t) - \frac{D}{2}\int_{\alpha}^{\pi/2}\sin\omega t \, d(\omega t)\right]$$

$$= \frac{4K}{\pi}\left[\frac{X}{2}\left[\omega t - \frac{\sin 2\omega t}{2}\right]_{\alpha}^{\pi/2} - \frac{D}{2}\left[-\cos\omega t\right]_{\alpha}^{\pi/2}\right]$$

$$= \frac{4K}{\pi}\left(\frac{X}{2}\left[\left(\frac{\pi}{2}-\alpha\right)-\frac{1}{2}(\sin\pi - \sin 2\alpha)\right] + \frac{D}{2}\left[\cos\omega t\right]_{\alpha}^{\pi/2}\right)$$

$\sin\alpha = \frac{D}{2X}$

$D = 2X\sin\alpha$

$$= \frac{4K}{\pi}\left[\frac{X}{2}\left[\left(\frac{\pi}{2}-\alpha\right)+\frac{\sin 2\alpha}{2}\right] + \frac{2X\sin\alpha}{2}\left(\cos\pi/2-\cos\alpha\right)\right]$$

$$= \frac{4K}{\pi}\left[\left(\frac{X}{2}\left(\frac{\pi}{2}-\alpha+\frac{\sin 2\alpha}{2}\right)\right)+\frac{\cancel{2}X\sin\alpha}{\cancel{2}}(-\cos\alpha)\right]$$

$$= \frac{4KX}{\pi}\left[\frac{\pi}{4}-\frac{\alpha}{2}+\frac{\sin 2\alpha}{4}-\sin\alpha\cos\alpha\right]$$

$$= \frac{4KX}{\pi}\left[\frac{\pi}{4}-\frac{\alpha}{2}+\frac{2\sin\alpha\cos\alpha}{4}-\sin\alpha\cos\alpha\right]$$

$$= \frac{4KX}{\pi}\left[\frac{\pi}{4}-\frac{\alpha}{2}-\frac{\sin\alpha\cos\alpha}{2}\right]$$

$$= \frac{4KX}{\pi}\left[\frac{\pi}{4}-\frac{1}{2}(\alpha+\sin\alpha\cos\alpha)\right]$$

$$= KX\left[\frac{4}{\pi}\times\frac{\pi}{4}-\frac{4}{\pi}\times\frac{1}{2}(\alpha+\sin\alpha\cos\alpha)\right]$$

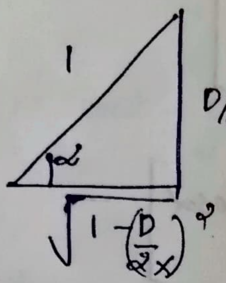
$$A_1 = KX\left[1-\frac{2}{\pi}(\alpha+\sin\alpha\cdot\cos\alpha)\right]$$

$$Y_1 = A_1 = KX\left[1-\frac{2}{\pi}(\alpha+\sin\alpha\cos\alpha)\right]$$

$$K_N(X,\omega) = \frac{Y_1}{X}\angle\phi_1 = \frac{KX}{X}\left[1-\frac{2}{\pi}(\alpha+\sin\alpha\cos\alpha)\right]$$

$$K_N(X,\omega) = K\left[1-\frac{2}{\pi}(\alpha+\sin\alpha\cos\alpha)\right]$$

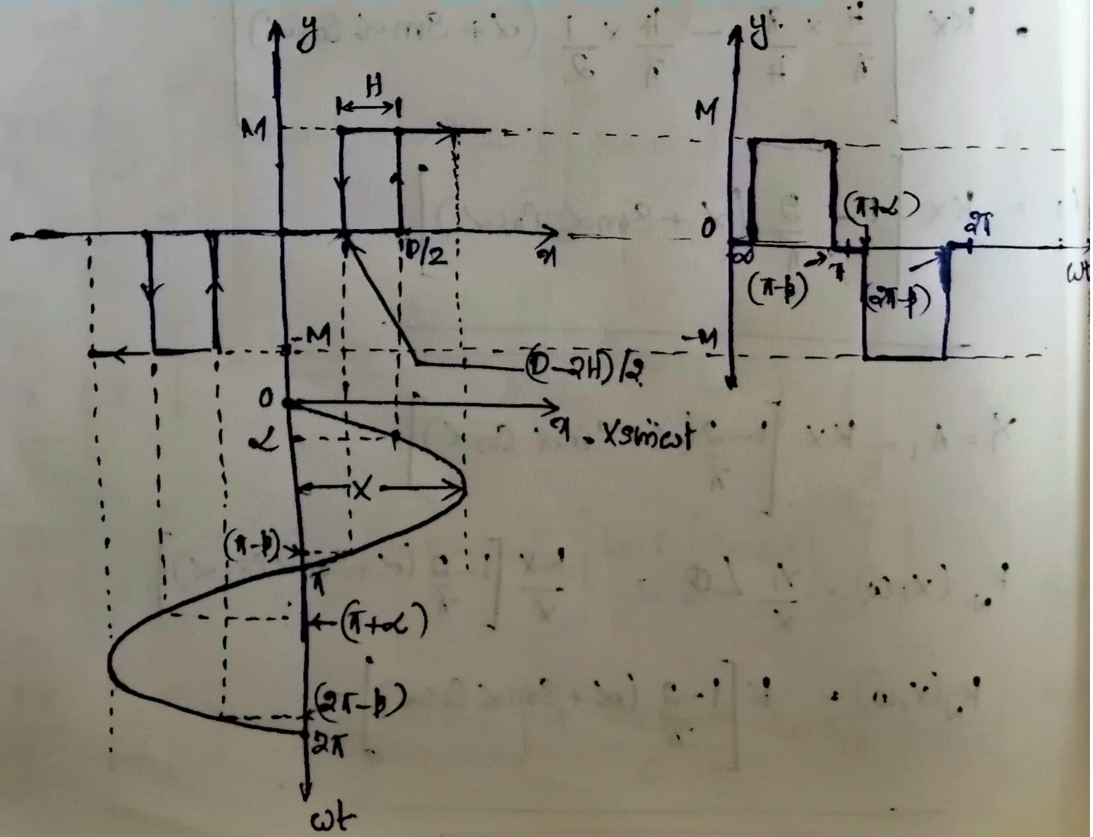
We've, $\sin\alpha = \frac{D}{2X}$



$\cos\alpha = \sqrt{1-\left(\frac{D}{2X}\right)^2}$

$$\therefore K_N(X,\omega) = K\,1 - \frac{2}{\pi}\left[\sin^{-1}\frac{D}{2X} + \frac{D}{2X}\sqrt{1-\left(\frac{D}{2X}\right)^2}\right], \text{ for } X > \frac{D}{2}$$

Describing func of relay with hysteresis & deadzone :



The i/p is $x = X \sin\omega t$ ——— ①

In fig, when $\omega t = \alpha$, $x = D/2$

$\therefore$ ① $\Rightarrow$ $D/2 = X \sin\alpha$

$$\sin\alpha = \frac{D}{2X}$$

$$\alpha = \sin^{-1}\frac{D}{2X}.$$

when $\omega t = \pi - \beta$, $x = \left(\frac{D}{2} - H\right)$.

① $\Rightarrow$ $\frac{D}{2} - H = X\sin(\pi - \beta)$

$$\frac{D}{2} - H = X \sin\beta.$$

$$\sin\beta = \frac{1}{X}\left(\frac{D}{2} - H\right) \quad \text{or} \quad \beta = \sin^{-1}\left[\frac{1}{X}\left(\frac{D}{2} - H\right)\right]$$

The o/p can be divided into 5 regions in a period of 2π and o/p equation for 5 regions are:

$$y = \begin{cases} 0; & 0 \le \omega t \le \alpha \\ M; & \alpha \le \omega t \le (\pi - \beta) \\ 0; & (\pi - \beta) \le \omega t \le (\pi + \alpha) \\ -M; & (\pi + \alpha) \le \omega t \le (2\pi - \beta) \\ 0; & (2\pi - \beta) \le \omega t \le 2\pi \end{cases}$$

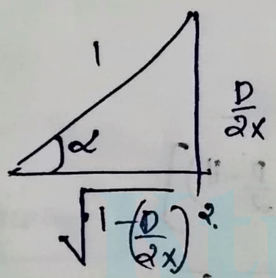
Describing func, $K_N(X, \omega) = \frac{Y_1}{X}\angle\phi_1$

$$A_1 = \frac{2}{\pi}\int_0^{\pi} y \sin\omega t \, d(\omega t) = \frac{2}{\pi}\int_{\alpha}^{\pi-\beta} y \sin\omega t \, d(\omega t).$$

$$= \frac{2M}{\pi}\left[-\cos\omega t\right]_{\alpha}^{\pi-\beta} = \frac{2M}{\pi} \; -\left[\cos(\pi-\beta) - \cos\alpha\right]$$

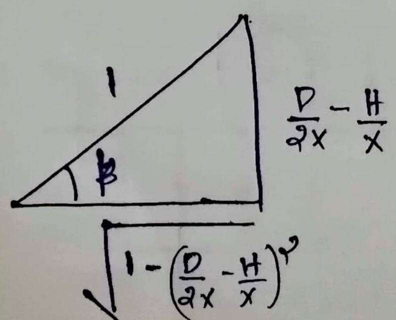
$$= \frac{2M}{\pi}\left(\cos\alpha + \cos\beta\right)$$

We've, $\sin\alpha = \frac{D}{2X}$



$$\therefore \cos\alpha = \sqrt{1-\left(\frac{D}{2X}\right)^2}.$$

also, $\sin\beta = \frac{D}{2X} - \frac{H}{X}$



$$\therefore \cos\beta = \sqrt{1-\left(\frac{D}{2X}-\frac{H}{X}\right)^2}$$

On substitution,

$$A_1 = \frac{2M}{\pi}\left[\sqrt{1-\left(\frac{D}{2X}\right)^2} + \sqrt{1-\left(\frac{D}{2X}-\frac{H}{X}\right)^2}\right].$$

$$B_1 = \frac{2}{\pi}\int_0^{\pi} y \cos\omega t \, d(\omega t) = \frac{2}{\pi}\int_{\alpha}^{\pi-\beta} M \cos\omega t \, d(\omega t)$$

$$= \frac{2M}{\pi}\left[\sin\omega t\right]_{\alpha}^{\pi-\beta} = \frac{2M}{\pi}\left(\sin(\pi-\beta) - \sin\alpha\right)$$

$$= \frac{2M}{\pi}\left(\sin\beta - \sin\alpha\right).$$

On substitution

$$B_1 = \frac{2M}{\pi}\left[\frac{D}{2X} - \frac{H}{X} - \frac{D}{2X}\right] = \frac{2M}{\pi}\left[-\frac{H}{X}\right]$$

$$\therefore Y_1 = \sqrt{A_1^2 + B_1^2}$$

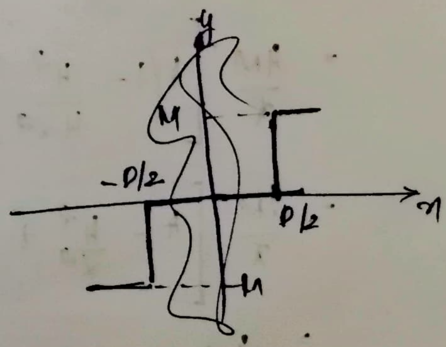
$$Y_1 = \left[\frac{4M^2}{\pi^2}\left[\sqrt{1-\left(\frac{D}{2X}\right)^2} + \sqrt{1-\left(\frac{D}{2X}-\frac{H}{X}\right)^2}\right]^2 + \frac{4M^2}{\pi^2}\left(\frac{H^2}{X^2}\right)\right]^{1/2}$$

$$\phi_1 = \tan^{-1}\frac{B_1}{A_1} = \tan^{-1}\frac{\frac{2M}{\pi}\left(-\frac{H}{X}\right)}{\frac{2M}{\pi}\left[\sqrt{1-\left(\frac{D}{2X}\right)^2} + \sqrt{1-\left(\frac{D}{2X}-\frac{H}{X}\right)^2}\right]}$$

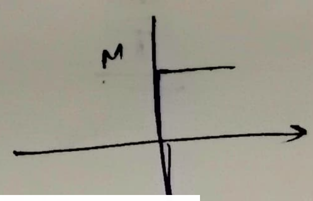
Ideal relay.

Here $D = H = 0$

$\therefore$ $Y_1 = \frac{2M}{\pi}$ and $\phi_1 = 0$



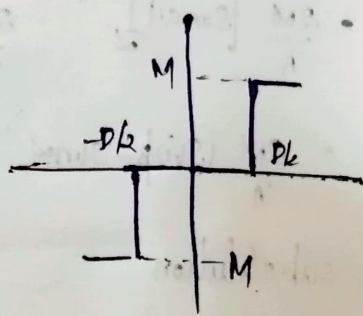
$$K_N(X,\omega) = \frac{2M}{\pi X}\angle 0^\circ$$



Relay with dead-zone:

Here $H = 0$

$$\therefore Y_1 = \left[\frac{4M^2}{\pi^2}\left\{2\sqrt{1-\left(\frac{D}{2x}\right)^2}\right\}^2\right]^{1/2}$$



$$= \frac{4M}{\pi}\sqrt{1-\left(\frac{D}{2x}\right)^2}$$

$$\phi_1 = 0$$

$$\therefore K_N(x,\omega) = \frac{Y_1}{X}\angle\phi_1 = \begin{cases} 0\,; & x < \frac{D}{2} \\ \frac{4M}{\pi}\sqrt{1-\left(\frac{D}{2x}\right)^2}\,; & x > \frac{D}{2} \end{cases}$$

Relay with hysteresis:

Here $D = H$

$$\therefore Y_1 = \left[\frac{4M^2}{\pi^2}\left\{\sqrt{1-\left(\frac{H}{2x}\right)^2} + \sqrt{1-\left(\frac{H}{2x}\right)^2}\right\}^2 + \frac{4M^2}{\pi^2}\left(\frac{H^2}{X^2}\right)\right]^{1/2}$$

$$= \frac{4M}{\pi}\left[4\left(1-\frac{H^2}{4x^2}\right) + \left(\frac{H^2}{X^2}\right)\right]^{1/2}$$

$$= \frac{2M}{\pi}\left[4 - \frac{H^2}{X^2} + \frac{H^2}{X^2}\right]^{1/2}$$

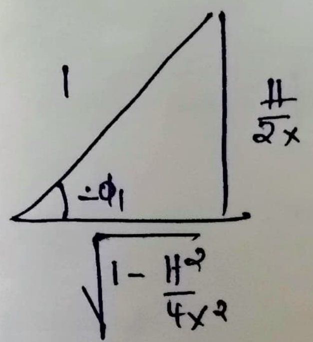
$$= \underline{\underline{\frac{4M}{\pi}}}$$

$$\phi_1 = \tan^{-1} \frac{\frac{2M}{\pi}\left(-\frac{H}{X}\right)}{\frac{2M}{\pi}\left[\sqrt{1-\left(\frac{H}{2X}\right)^2}+\sqrt{1-\left(\frac{-H}{2X}\right)^2}\right]} = \tan^{-1} \frac{-H/X}{2\sqrt{1-\frac{H^2}{4X^2}}}$$

$$= -\tan^{-1} \frac{H/2X}{\sqrt{1-\frac{H^2}{4X^2}}}$$

$$-\phi_1 = \tan^{-1} \frac{H/2X}{\sqrt{1-\frac{H^2}{4X^2}}}$$

$$\tan(-\phi_1) = \frac{H/2X}{\sqrt{1-\frac{H^2}{4X^2}}}$$



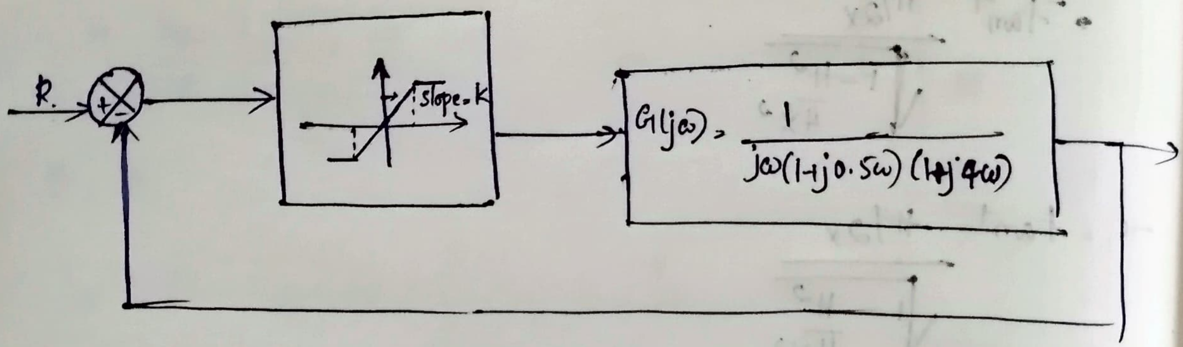
$$\therefore \sin(-\phi_1) = \frac{H}{2X}.$$

$$-\phi_1 = \sin^{-1}\left(\frac{H}{2X}\right)$$

$$\text{or } \phi_1 = -\sin^{-1}\left(\frac{H}{2X}\right).$$

$$K_N(X,\omega) = \frac{Y_1}{X}\angle\phi_1 = \begin{cases} 0 ; & X < H/2 \\ \frac{4M}{\pi X}\angle -\sin^{-1}\left(\frac{H}{2X}\right) ; & X > \frac{H}{2} \end{cases}$$

5) Consider a unity feedback s/m ~~with~~ having saturation amplifier with gain K. Determine the max. value of K for the s/m to stay stable. what would be the freq. and nature of minute cycle for a gain, K = 2.5



Polar plot of $G(j\omega)$ at K=1,

$$G(j\omega) = \frac{K}{j\omega(1+j0.5\omega)(1+j4\omega)}$$

Let K = 1, $G(j\omega) = \frac{1}{j\omega(1+j0.5\omega)(1+j4\omega)}$.

$$G(j\omega) = \frac{1}{(\omega\angle 90^\circ)\left(\sqrt{1+(0.5\omega)^2}\angle \tan^{-1}0.5\omega\right)\left(\sqrt{1+(4\omega)^2}\angle \tan^{-1}4\omega\right)}$$

$$|G(j\omega)| = \frac{1}{\omega\sqrt{1+(0.5\omega)^2}\cdot\sqrt{1+(4\omega)^2}} \; ; \; \angle G(j\omega) = -90^\circ - \tan^{-1}0.5\omega - \tan^{-1}4\omega$$

ω (rad/sec)	0.4	0.5	0.6	0.8	1.0	1.2
$\lvert G(j\omega)\rvert$	1.299	0.867	0.613	0.346	0.216	0.145
$\angle G(j\omega)$	-159°	-167°	-174°	-184°	-192°	-199°
$G_R(j\omega)$	-1.212	-0.844 ≈ 0.85	-0.609 ≈ -0.61	-0.345	-0.211	$-0.137 = -0.14$
$G_{im}(j\omega)$	-0.465 $\approx (-0.47)$	-0.195 (-0.2)	-0.064 ≈ -0.06	0.024	0.044	0.047 ≈ 0.05

Polar plot of G(jω) when K = 2.5:

$$G(j\omega) = \frac{2.5}{j\omega(1+j0.5\omega)(1+j4\omega)}$$

$$|G(j\omega)| = \frac{2.5}{\omega\sqrt{1+(0.5\omega)^2}\cdot\sqrt{1+(4\omega)^2}} \; ; \; \angle G(j\omega) = -90^\circ - \tan^{-1} 0.5\omega - \tan^{-1} 4\omega$$

ω (rad/sec)	0.6	0.65	~~0.6~~ 0.75	0.8	1.0	1.2
$\lvert G(j\omega)\rvert$	1.535 ~~3.246~~	1.313 ~~2.164~~	0.981	0.865	0.54	0.363
$\angle G(j\omega)$	-174	-177	-182	-184	-192	-199°
$G_R(j\omega)$	-1.52	-1.31	-0.99	-0.87	-0.53	-0.34
$G_{im}(j\omega)$	-0.16	-0.07	0.03	0.06	0.11	0.12

$$K_N = \begin{cases} 1, & x < s. \\ \frac{2K}{\pi}\left(\beta + \sin\beta\cos\beta\right), & x > s \end{cases}$$

for $k=1, s=1$

$$-1/K_N = \begin{cases} 1\angle -180^\circ, & x < 1 \\ \frac{\pi}{2(\beta + \sin\beta\cos\beta)}, & x > 1 \end{cases}$$

To find the max. value of K for stability:

Case 1: when $K=1$

when $K=1$ $G(j\omega)$ locus does not enclose $-1/K_N$. Hence the s/m is stable and $G(j\omega)$ crosses -ve real axis at $-1+j0$.

$$\therefore G(j\omega) = -1 = 1\angle -180^\circ$$

$$\therefore |G(j\omega)| = 1, \quad \angle G(j\omega) = -180^\circ.$$

At $\omega = \omega_{p1}$,

$$\angle G(j\omega) = -90^\circ - \tan^{-1} 0.5\omega_{p1} - \tan^{-1} 4\omega_{p1}$$

$$= -180^\circ.$$

$\tan^{-1} 0.5\omega_{f1} + \tan^{-1} 4\omega_{f1} = 90°$

$\tan(\tan^{-1} 0.5\omega_{f1} + \tan^{-1} 4\omega_{f1}) = \tan 90°$

$$\frac{0.5\omega_{f1} + 4\omega_{f1}}{1 - 0.5\omega_{f1} \times 4\omega_{f1}} = \infty$$

$\Rightarrow \quad 1 - 0.5\omega_{f1} \times 4\omega_{f} = 0$

$2\omega_{f1}^2 = 1$

$\omega_{f1} = \sqrt{0.5}$ rad/sec or $\omega_{f1} = 1/\sqrt{2}$ rad/s

we've, $|G(j\omega)| = 1$

$$\Rightarrow \frac{K}{\omega_{f1}\sqrt{1+(0.5\omega_{f1})^2} \cdot \sqrt{1+(4\omega_{f1})^2}} = 1$$

$$\frac{K}{\frac{1}{\sqrt{2}}\left[1+\left(\frac{0.5}{\sqrt{2}}\right)^2\right]^{1/2} \cdot \left[1+\left(\frac{4}{\sqrt{2}}\right)^2\right]^{1/2}} = 1$$

$$\frac{4K}{9} = 1$$

$$K = \frac{9}{4} = 2.25$$

For $K < 2.25$, the s/m is stable.

Case 2: when $K = 2.5$

when $K = 2.5$, $G(j\omega)$ locus intersects $-1/K_N$ at $-1.11 + j0$.

$G(j\omega) = +1.11 \angle -180^\circ$

At $\omega = \omega_{p2}$

$$\angle G(j\omega) = -90^\circ - \tan^{-1} 0.5\omega_{p2} - \tan^{-1} 4\omega_{p2} = -180^\circ$$

$$\tan^{-1} 0.5\omega_{p2} + \tan^{-1} 4\omega_{p2} = 90^\circ$$

$$\tan(\tan^{-1} 0.5\omega_{p2} + \tan^{-1} 4\omega_{p2}) = \tan 90^\circ$$

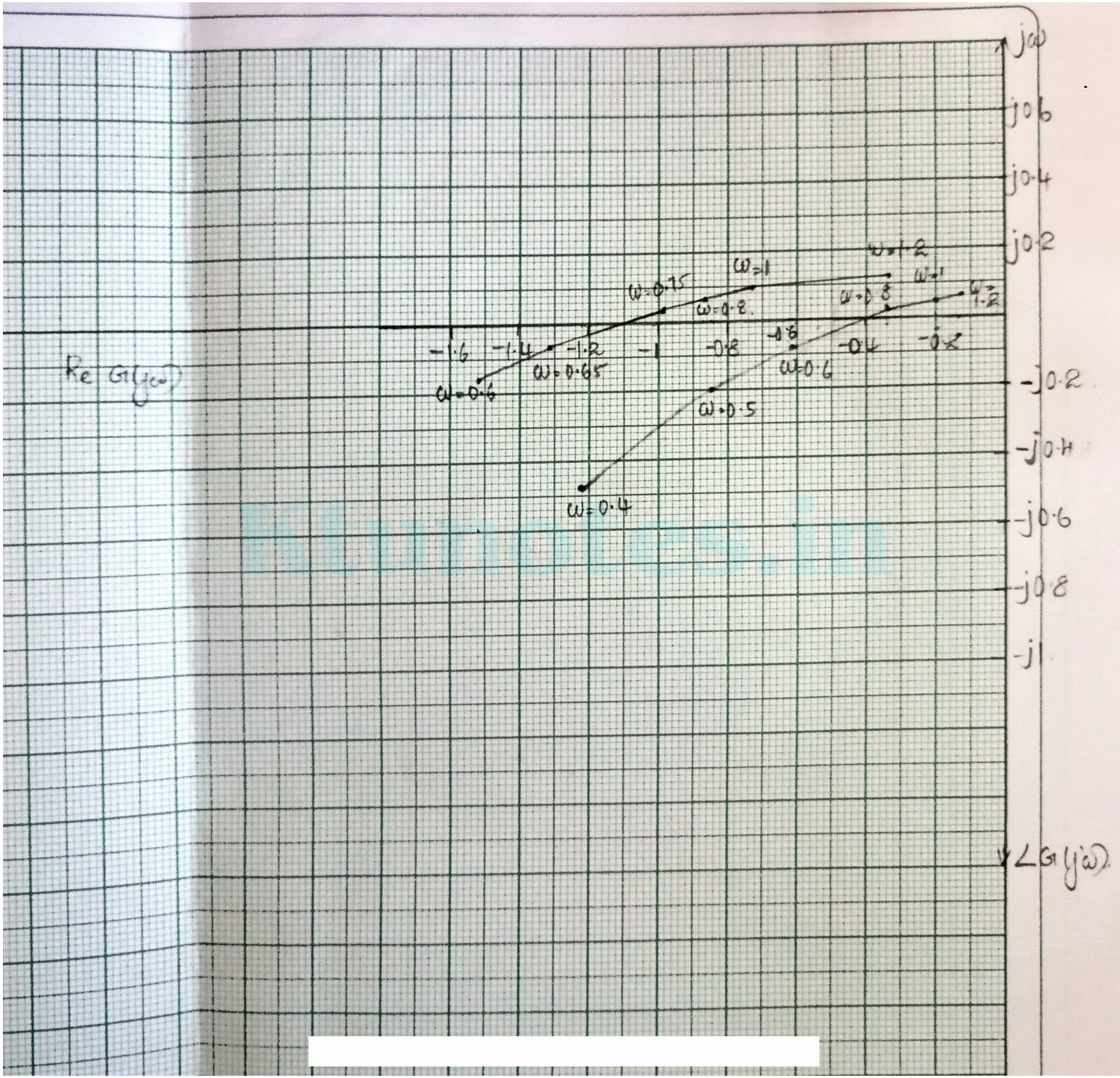
$$\frac{0.5\omega_{p2} + 4\omega_{p2}}{1 - 0.5\omega_{p2} \cdot 0.4\omega_{p2}} = \infty$$

$$1 - 0.5\omega_{p2} \cdot 0.4\omega_{p2} = 0.$$

$$\omega_{p2} = 1/\sqrt{2}$$ $\therefore$ freq. of minute cycle $= 1/\sqrt{2}$ at $K = 2.5$

We've, $|G(j\omega)| = 1.11$

$$\Rightarrow \frac{K}{\omega_{p2} \cdot \sqrt{1 + (0.5\omega_{p2})^2} \sqrt{1 + (4\omega_{p2})^2}} = 1.11$$


jω
j0.6
j0.4
j0.2
-j0.2
-j0.4
-j0.6
-j0.8
-j1
∠G(jω)
Re G(jω)
-1.6
-1.4
-1.2
-1
-0.8
-0.6
-0.4
-0.2
ω=0.6
ω=0.65
ω=0.75
ω=0.8
ω=1
ω=1.2
ω=0.4
ω=0.5
ω=0.6
ω=0.8
ω=1
ω=1.2

Result:

1) At $k=1$, s/m is stable

2) S/m remains stable for $k < 2.25$

3) $k = 2.5$, limit cycle occur, freq. of limit cycle $= 1/\sqrt{2}$

3rd module & 4th:

- pole placement technique (problm)
- controll. observability problm
- describing func. of saturation
- " " relay with hysteresis & deadzone.
- significance of PBH test controll. & observ. (with eg for 10 marks)
- notes on reduced state observer. (and problm).
- duality principle for controll.